

# IndiLoRa X Single-Channel Indoor LoRaWAN® Gateway Datasheet

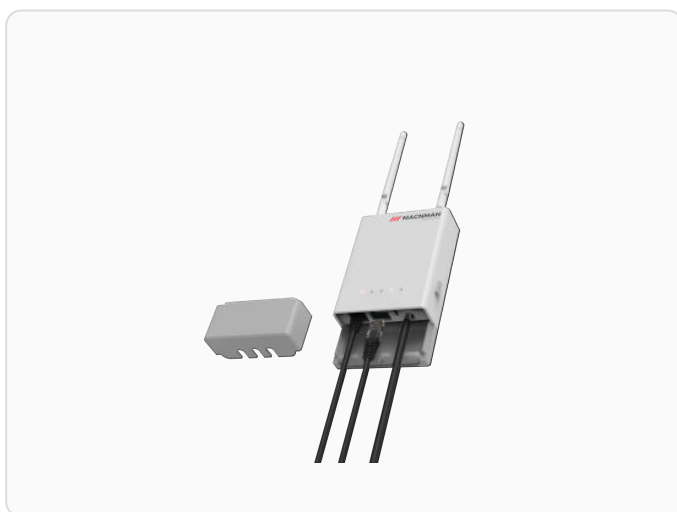
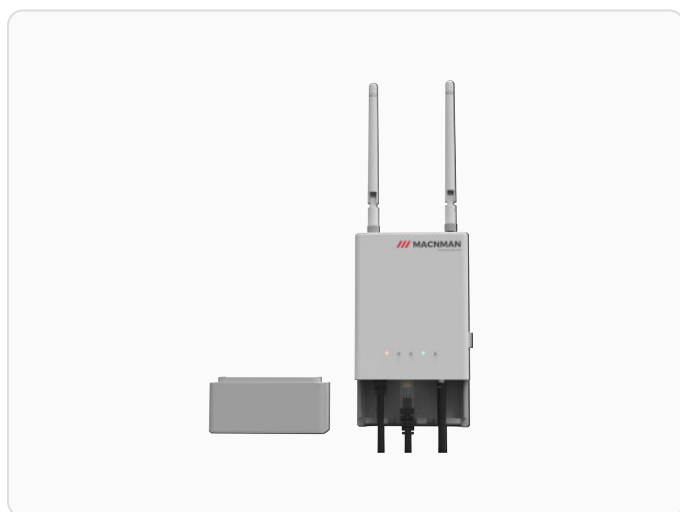
## Product Overview

**IndiLoRa X1 is an ultra-affordable single-channel indoor LoRaWAN® gateway** designed for small-scale IoT deployments and legacy industrial system upgrades. Built on the powerful nRF5340 dual-core MCU, it bridges LoRa® sensors with both cloud platforms and local RS-485 networks simultaneously — enabling seamless modernization of existing PLCs, meters, and building automation systems.

With Wi-Fi backhaul, IndiLoRa X1 eliminates the need for Ethernet cabling, making installation fast and extremely cost-efficient in retail stores, warehouses, apartments, hospitals, small factories, and utility rooms. The fanless design ensures silent operation and support for 24×7 monitoring applications.

Supporting AES-128 encrypted communication, OTA updates, **MQTT/HTTPS cloud delivery**, and **direct RS-485 Modbus output**, IndiLoRa X1 allows rapid integration into SCADA, BMS, EMS, utility monitoring and more — without replacing existing equipment.

Whether enabling indoor environmental monitoring, asset tracking, energy meters, or low-density sensor networks — IndiLoRa X1 delivers a budget-friendly indoor LoRa® gateway solution that bridges Industry 3.0 with Industry 4.0.





## Key Features

### Ultra-Low Cost LoRa® Gateway

Designed for cost-sensitive IoT installations with compact coverage needs in smart buildings and commercial environments.

### RS-485 Modbus Output Support

Direct interface with legacy PLCs, meters and BMS systems for simultaneous LoRa-to-wired integration.

### Focused Single-Channel LoRa® Operation

Ideal for low-density sensor networks inside controlled indoor areas.

### Plug-and-Play Deployment

Zero-touch provisioning using MacCloud for rapid installation by onsite technicians.

### Wi-Fi Backhaul Connectivity

Cloud communication using Wi-Fi no Ethernet wiring required for installation.

### Powered by nRF5340 MCU

Dual-core processing with secure and power-efficient wireless communication.

### Secure Communication

AES-128 data encryption and TLS-based secure OTA firmware upgrades.

### Quiet & Low-Power Design

Fanless architecture built for 24x7 uptime in noise-sensitive environments.

## Suited Applications

### Small Smart Building Deployments

Cost-efficient indoor LoRa® coverage for energy monitoring, room comfort sensors and building automation in compact areas.

### Warehouse & Storage Spaces

Ideal for indoor asset tracking, stock environment monitoring and automated supply chain alerts.

### Commercial Retail Outlets

Enables IoT connectivity for shelf monitoring, occupancy analytics and refrigeration compliance without wired infrastructure.

### Hospitals & Healthcare Rooms

Adds secure connectivity for patient safety, air quality monitoring and equipment status without major installation work.

### Apartments & Residential Smart Homes

Facilitates LoRa®-based water meters, energy monitoring and safety sensors for housing automation.

### Hotels & Hospitality Spaces

Supports smart room solutions and maintenance alert systems without disturbing infrastructure aesthetics.

### Legacy Industrial Machine Rooms

Provides RS-485 integration for PLCs and meters, bridging Industry 3.0 devices with modern IoT cloud systems.

### Data Center & IT Racks

Perfect for localized monitoring of temperature, humidity and access conditions inside server environments.

## Technical Specifications

### General Specifications

|                   |                              |
|-------------------|------------------------------|
| Part Order Number | IndiLoRa X                   |
| Product Brand     | Macnman Technologies Pvt Ltd |

### Hardware Specifications

|                      |                                                    |
|----------------------|----------------------------------------------------|
| Processor Type       | nRF5340 Dual-Core Bluetooth® LE & LoRa Gateway MCU |
| Number of Cores      | Dual-Core (1× Application + 1× Network Core)       |
| Clock Speed          | 128 MHz (Application Core) / 64 MHz (Network Core) |
| Memory Size          | 512 KB RAM                                         |
| Memory Architecture  | Low-Power Multi-Core Shared Memory Architecture    |
| Internal Flash       | 1 MB On-Chip Flash + External Flash Support        |
| USB Port             | Debug Interface (Bottom of Enclosure)              |
| Connector Type       | USB Type-C                                         |
| Interface Protection | Surge Protection Enabled                           |

### LoRa Specifications

|                           |                                                      |
|---------------------------|------------------------------------------------------|
| Compatibility             | Macnman LoRa Nodes & Controllers                     |
| Supported Frequency Bands | IN865                                                |
| Channel Support           | Single-Channel (1 × Uplink / 1 × Downlink)           |
| Data-Rate                 | Fixed Data-Rate – No ADR (single-channel limitation) |

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| Tx Power        | +23 dBm @ IN865/868 MHz +20 dBm @ 915 MHz                            |
| Rx Sensitivity  | Down to –136 dBm @ SF12, BW 125 kHz                                  |
| Antenna Type    | External High-Efficiency LoRa Antenna                                |
| Device Capacity | Optimized for Localized Indoor Installations with Small Node Density |

### Native RS485 Output

|                      |                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------|
| Interface Type       | RS-485 Half-Duplex                                                                         |
| Protocol             | Modbus RTU (Slave Mode)                                                                    |
| Baud Rates Supported | 9600 / 19200 / 38400 / 57600 / 115200 bps (Configurable)                                   |
| Data Format          | 8-N-1 (Default) / Configurable                                                             |
| Command Processing   | Receives LoRa® sensor data and responds to host read commands via Modbus Holding Registers |
| Isolation            | Built-in surge and ESD protection for industrial environments                              |
| Connector            | 2-Pin Terminal Block                                                                       |
| Integration          | Compatible with Industrial PLCs, BMS, HVAC Controllers and SCADA systems                   |

### Backhual WiFi

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| Bands              | 2.4 GHz ISM Band<br>5 GHz ISM Band                            |
| Standards          | IEEE 802.11 b/g/n, 802.3, 802.3u, 802.11e-compatible bursting |
| Modulation Schemes | BPSK, QPSK, 16 QAM, 64 QAM                                    |
| Security           | WEP, TKIP, AES, WPA, WPA2                                     |
| Antenna            | External Antenna Integrated in Enclosure                      |

### Backhual Cellular Lte\*

|                    |                                                                |
|--------------------|----------------------------------------------------------------|
| Cellular Bands     | B1 / B3 / B5 / B7 / B8 / B20 / B28                             |
| Standards          | 3GPP LTE, IEEE 802.3, IEEE 802.3u, 802.11e-compatible bursting |
| Modulation Schemes | BPSK, QPSK, 16-QAM, 64-QAM                                     |
| Security           | WEP, TKIP, AES, WPA, WPA2                                      |

|         |                                          |
|---------|------------------------------------------|
| Antenna | Internal Antenna Integrated in Enclosure |
|---------|------------------------------------------|

## Supported Protocols

|                     |                       |
|---------------------|-----------------------|
| Supported Protocols | MQTT<br>HTTP<br>HTTPS |
|---------------------|-----------------------|

## Physical Characteristics

|                            |                                |
|----------------------------|--------------------------------|
| Power Supply               | 9-36 V PoE                     |
| Operating Temperature      | -40°C to +80°C                 |
| Humidity Range             | 0% to 100% RH (non-condensing) |
| Ingress Protection         | IP35                           |
| Housing Material           | Aluminium                      |
| Color                      | Matte White                    |
| Weight                     | ~120g                          |
| Mounting Options(Brackets) | Pole Mounting / Wall Mounting  |

## Certifications & Compliance

|      |                            |
|------|----------------------------|
| WPC  | Compliant                  |
| RoHS | Conforms to RoHS directive |

## Security & Encryption

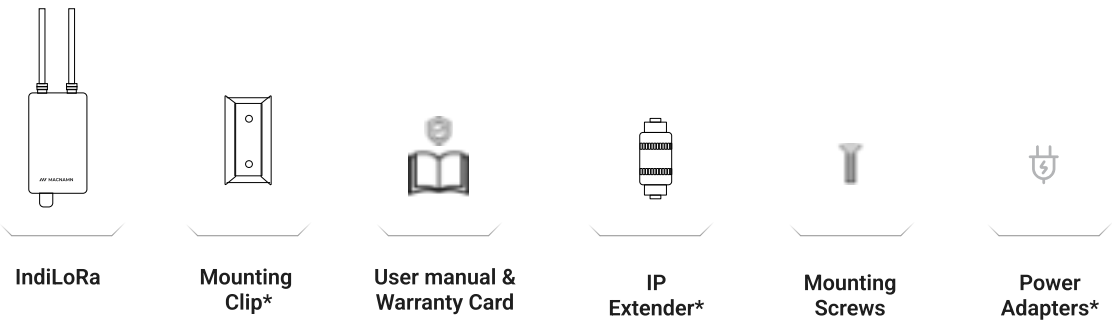
|                     |                                    |
|---------------------|------------------------------------|
| Data Encryption     | AES-128 end-to-end encryption      |
| Authentication      | OTAA activation modes (Nodes)      |
| Firmware Protection | Secure boot and signed OTA updates |

# IndiLoRa X Variants

| VARIANT NAME | POWER SOURCE       | WIFI | ETHERNET | CELLULAR | IDEAL FOR                                                                                                                                                   |
|--------------|--------------------|------|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IndiLoRa X1  | 12–24 V<br>(Power) |      |          |          | Ideal for small indoor networks with stable Wi-Fi backhaul such as offices, retail stores and hospitality environments.                                     |
| IndiLoRa X2  | 12–24 V<br>(Power) |      |          |          | Designed for mission-critical deployments where Ethernet, Wi-Fi and LTE failover ensure uninterrupted connectivity in industrial and multi-site facilities. |

## What's in the Box?

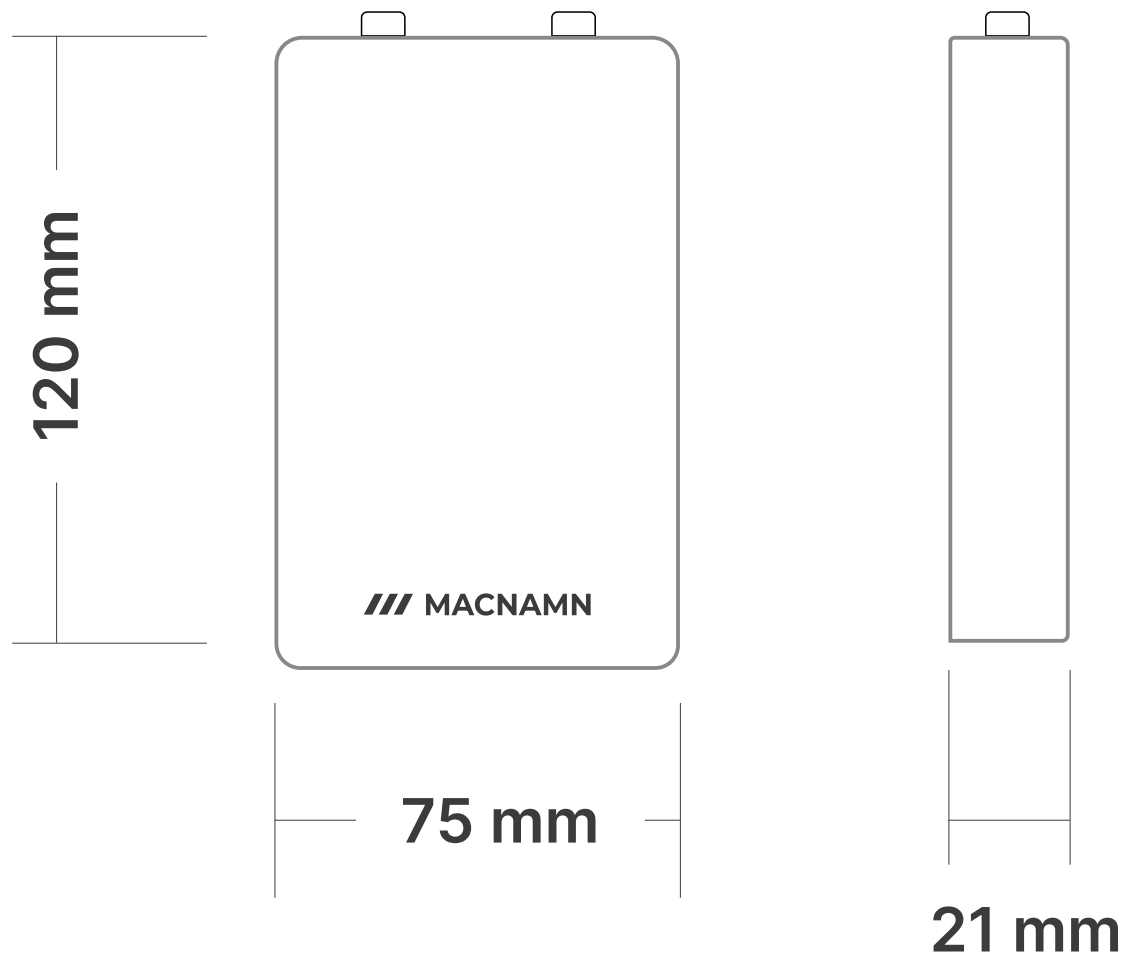
Make sure box contains following items before installations



**Note**

Accessories must be purchased separately & may not be included in the box.

## IndiLoRa X Dimensions



## Frequently Asked Questions

### What is the IndiLoRa X used for?

IndiLoRa X is an ultra-affordable single-channel indoor LoRaWAN gateway for small-scale IoT deployments and legacy industrial system upgrades. Typical applications include Small Smart Building Deployments, Commercial Retail Outlets, Warehouse & Storage Spaces, Hospitals & Healthcare Rooms, Apartments & Residential Smart Homes and Legacy Industrial Machine Rooms.

### **Which LoRa channels and frequency bands does the IndiLoRa X support?**

The IndiLoRa X is a single-channel gateway: Single-Channel (1 × Uplink / 1 × Downlink). Supported frequency band: IN865. It works with Macnman LoRa Nodes & Controllers. Receive sensitivity: Down to –136 dBm @ SF12, BW 125 kHz.

### **Which backhaul options does the IndiLoRa X have?**

The IndiLoRa X offers Wi-Fi and cellular LTE (optional) backhaul. Cellular bands: B1 / B3 / B5 / B7 / B8 / B20 / B28.

### **How is the IndiLoRa X powered?**

The IndiLoRa X is powered from 9-36 V PoE.

### **What is the IP rating and operating temperature of the IndiLoRa X?**

The IndiLoRa X has an ingress protection rating of IP35 and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

## **Related Products and Resources**

- Related datasheets: [Setu SCX single-channel indoor LoRa gateway](#), [MacLink Gen-2 industrial outdoor LoRaWAN gateway](#), [Setu LX Series indoor LoRaWAN gateway](#), [FireGaurd Gen 1 fire equipment health monitoring controller](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)